DAY 33 | POST-READ

LLM Deployment: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
Architectures	Hosted API, cloud-hosted in your tenant, self-hosted open model. Gateway for shared control.
AI gateway	One entry point: auth, limits, budgets, guardrails, logging, provider switching.
Environments	Dev, test, prod. Same image and manifest; only configuration changes.
Secrets	From environment or Key Vault; fail fast if missing in production.
Inference pipeline	Validate, enrich, generate, check, respond; time every stage.
Async + semaphore	Serve many requests while waiting; cap calls in flight to avoid 429s.
Streaming	Lower time to first token; StreamingResponse in FastAPI.
Queues	Long jobs handled by workers; the client checks back or gets a callback.
Rollouts	Canary, blue-green, shadow, feature flags.

Part B: Going beyond today

1. Self-hosting open models

Inference servers such as vLLM serve open models efficiently with continuous batching. Self-hosting makes sense with strict data rules, very high volume, or a need for custom models; budget for GPU capacity and operations.

2. Server-sent events

For browser chat UIs, servers often stream with server-sent events (text/event-stream), which browsers read easily and can reconnect automatically.

3. Back-pressure

When more requests arrive than you can serve, decide on purpose: queue them, return 429 with Retry-After, or degrade to a cheaper model. Never let them pile up silently until everything times out.

4. Regional deployment

For users in India and Japan, choose regions close to them and check that the models you need are available there. Data residency rules may decide the region for you.

5. Rollback criteria

Decide before a canary starts which numbers trigger a rollback: error rate, p95 latency, guardrail blocks, thumbs-down rate. Automate the rollback if you can.

Part C: Check yourself

- When would you self-host an open model?
- What does an AI gateway centralise?
- Why fail fast when a secret is missing in production?
- What problem does a semaphore solve?
- Compare canary and blue-green releases.

Part D: Practice ideas for tonight

- Add a /health and /ready endpoint to the notebook's service and a readiness check for the API key.
- Add a feature flag that turns streaming on for one user group.
- Write rollback criteria for your capstone's first release.

Coming up on Day 34: observability: structured logging, tracing, quality and latency monitoring, drift detection and feedback loops.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today